RoboPIN: Grounded Embodied Reasoning via Pinned Chain-of-Thought

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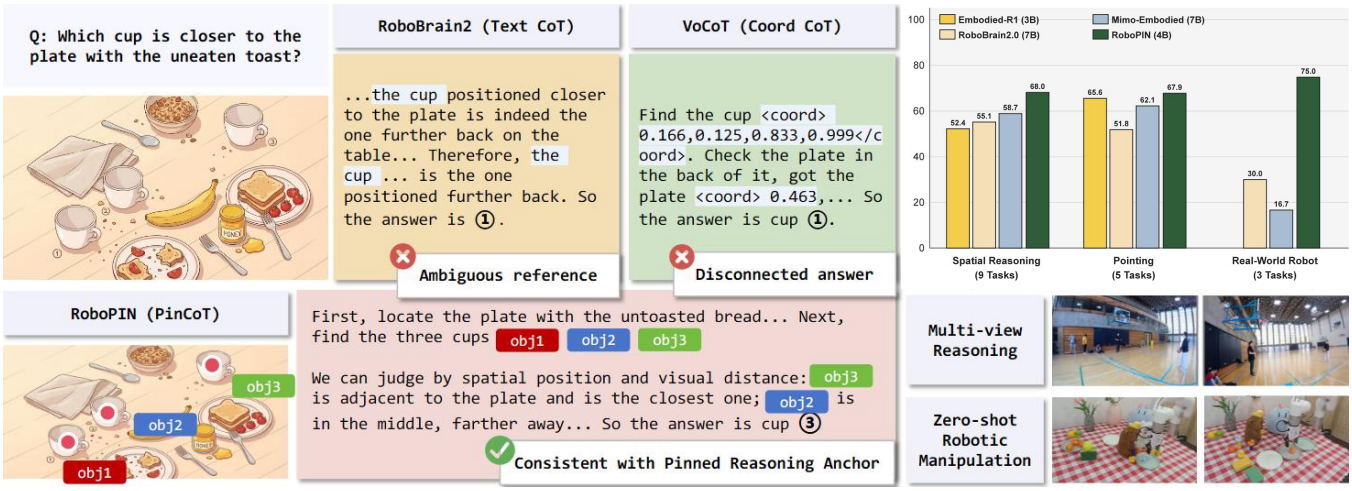


Figure 1: Comparison of three chain-of-thought paradigms for embodied reasoning. Text CoT relies on ambiguous textual entity references. Coord CoT embeds coordinates but lacks identity tracking, causing reasoning-answer disconnection. PinCoT (ours) pins each reasoning step to visual evidence, ensuring identity-level traceability to the final answer.

Abstract

Embodied reasoning requires models to perceive task-relevant objects and spaces in physical environments and maintain consistent visual grounding throughout multi-step reasoning. However, current vision-language models rely on text-only or coordinate-augmented chain-of-thought, where entity references remain implicit and ambiguous. This may cause the reasoning process to decouple from visual evidence, entity references to drift across steps, and a causal disconnection between the reasoning trajectory and the final answer, with these problems further amplified in multi-view scenarios due to cross-view appearance changes. To address these issues, we propose Pinned Chain-of-Thought (PinCoT), a structured reasoning paradigm that pins every reasoning step to visual evidence. PinCoT introduces the concept of reasoning anchor, which binds each task-relevant entity to a structured visual anchor with entity name, unique identity, view index, and

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spatial grounding, enabling consistent entity tracking across reasoning steps and views. We build a fully automated data generation pipeline to construct PIN-170K, a high-quality PinCoT-formatted reasoning dataset. We then train RoboPIN through three-stage post-training that progressively injects embodied knowledge, structured reasoning ability, and process-supervised alignment, with rewards that directly constrain both anchor localization and identity consistency during reasoning. On 14 benchmarks covering embodied spatial reasoning, multi-view reasoning, and pointing, RoboPIN with only 4B parameters surpasses 7B level open-source embodied models on average, achieving a 12% average improvement over the strongest 7B baseline, Mimo-Embodied. Further analysis shows that PinCoT improves grounding accuracy and cross-step identity consistency, validating the effectiveness of process supervision.

CCS Concepts

• **Computing methodologies** → *Scene understanding*; **Spatial and physical reasoning**.

Keywords

Embodied Reasoning, Spatial Reasoning, Multi-view Reasoning

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1 Introduction

Embodied reasoning is a core capability for physical-world intelligent systems [4, 30, 37, 41, 50]. Tasks such as spatial reasoning, fine-grained pointing, and long-horizon planning require models to perceive task-relevant objects and spatial regions in complex environments, while maintaining tight coupling between reasoning steps and visual evidence throughout multi-step inference [2, 5, 6, 34, 46].

Despite the necessity of this tight coupling, current Vision-Language Models (VLMs) struggle to maintain it in embodied tasks. [10, 13, 34]. Although chain-of-thought reasoning has evolved from text to multimodal and visually grounded forms [22, 53], existing methods still lack reasoning consistency. Multimodal chain-of-thought methods [21, 31, 49, 54] incorporate visual information into reasoning, but reasoning steps remain largely textual and lack explicit spatial grounding. Coordinate-augmented approaches [3, 15, 17] go further by embedding coordinates or bounding boxes into reasoning chains, yet these spatial references are isolated and lack explicit identity tracking across steps. As shown in Figure 1, this leads to two core problems. First, **reasoning decouples from visual evidence**: entity references are implicit textual descriptions that cannot ensure each step looks at the correct target, and cross-step entity references are prone to drift, especially when multiple same-category instances exist. Second, **reasoning-to-answer misalignment**: when intermediate steps contain reference drift, the final answer loses reliable visual grounding. Consequently, a model might correctly analyze object A initially, but silently shift its attention and point to object B in the end. Current paradigms fail to detect this disconnection because the underlying causal chain is broken. These issues are further amplified in multi-view settings, these problems are further amplified because the same object undergoes appearance changes across viewpoints [9, 12, 40, 44]. We unify these issues as the *reasoning consistency* problem in embodied reasoning, spanning cross-step and cross-view consistency. The root cause is the lack of a unified *reasoning anchor* mechanism to persistently bind visual evidence throughout reasoning.

To address these issues, we propose *Pinned Chain-of-Thought* (*PinCoT*), a structured reasoning paradigm that "pins" every reasoning step onto visual evidence. The core of PinCoT is the *reasoning anchor*: each task-relevant entity is bound to a structured visual anchor with a name, unique identity, view index, and spatial grounding. During reasoning, an entity is introduced with a full tag establishing its localization and identity upon first appearance; subsequent steps refer to the same entity via its ID. This enables identity-consistent reasoning, which extends to multi-view settings where the same object shares its ID across different viewpoints. Unlike coordinate-embedding methods such as Shikra [3], region-grounding approaches such as Ferret [45], and visual chain-of-thought methods such as VoCoT [15], PinCoT does not perform static grounding external to the reasoning process, but rather treats these visual references as persistently trackable anchors within the reasoning chain. To obtain supervision in the PinCoT format [31, 51], we build a fully automated pipeline and use it to construct the PIN-170K reasoning dataset, as shown in Figure 2. The pipeline transforms multimodal data into PinCoT-format reasoning supervision through three stages: semantic parsing, precise point grounding, and reasoning chain generation, incorporating dedicated quality assurance throughout the process.

Building on this framework and data, we train RoboPIN through SFT, CoT-SFT, and RFT three-stage post-training, which respectively inject embodied knowledge, structured reasoning ability, and process-supervised alignment [48]. The RFT stage employs a composite reward function that simultaneously supervises the reasoning process and the final answer [18], effectively mitigating reference drift across reasoning steps.

Extensive experiments show that RoboPIN achieves best average performance across 14 benchmarks covering embodied spatial reasoning, multi-view reasoning, and pointing tasks [10, 12, 13, 29, 34, 40]. Further analysis confirms that PinCoT substantially improves the localization accuracy of reasoning anchors and cross-step identity consistency, validating the effectiveness of process supervision on reasoning quality. The main contributions are as follows:

- We propose **PinCoT**, introducing the reasoning anchor concept that converts implicit visual attention during reasoning into explicit, trackable structured visual anchors, achieving cross-step and cross-view reasoning consistency.
- We build a fully automated, high-quality data generation pipeline and construct the PIN-170K dataset, enabling large-scale generation of PinCoT-format reasoning supervision.
- We train **RoboPIN** with three-stage post-training and process-supervised rewards that jointly optimize reasoning process quality and answer correctness. Impressively, with only 4B parameters, RoboPIN achieves best average performance across 14 benchmarks spanning embodied spatial reasoning, pointing, and multi-view reasoning, outperforming 7B level open-source embodied

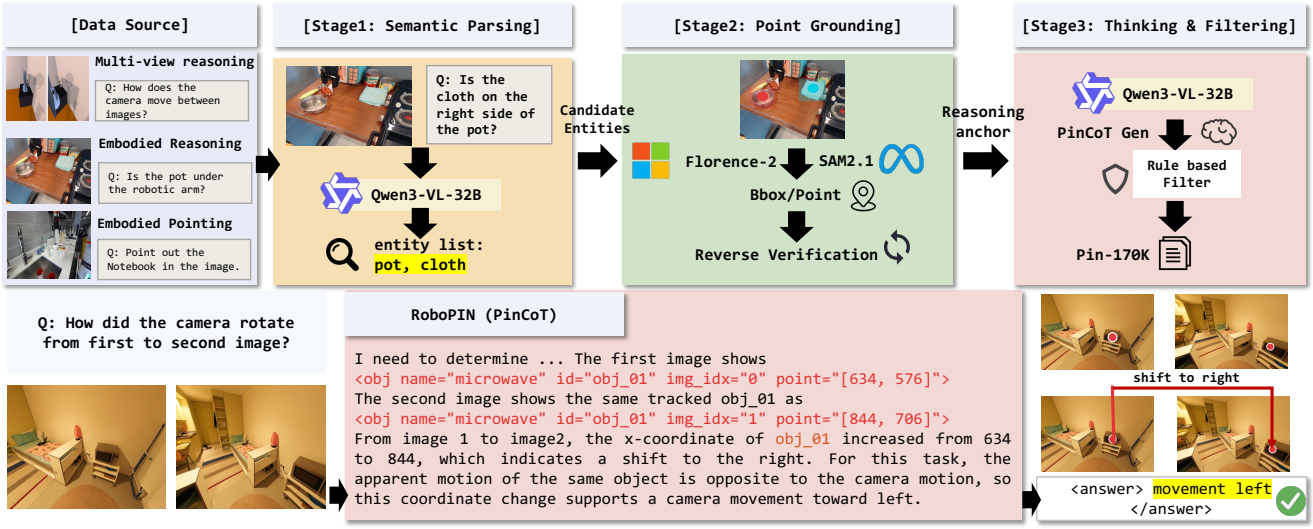


Figure 2: Overview of RoboPIN. Our framework enables grounded embodied reasoning through PinCoT, which pins every reasoning step onto visual evidence with identity-tracked anchors.

models with a 12% average improvement over the strongest 7B baseline Mimo-Embodied [14].

2 Related Work

2.1 Embodied Spatial Reasoning

Spatial understanding is essential for embodied agents to bridge the "seeing-to-doing" gap. [48]. To act reliably in physical environments, a model must not only recognize objects, but also infer their locations, spatial relations, and task-relevant regions. Recent vision-language models have substantially improved this capability by incorporating explicit spatial coordinates, geometric reasoning, and scene-structure modeling into multimodal representations [2, 5]. In robotic manipulation settings, another important line of work predicts actionable points and visual traces that can directly support downstream control, such as grasping, placing, or interaction planning [27, 46, 47]. Together, these efforts have significantly strengthened embodied models in tasks such as spatial referring, target localization, and manipulation-oriented grounding.

2.2 Visually Grounded Chain-of-Thought

Chain-of-Thought (CoT) [42] significantly enhances the reasoning abilities of large language models. To extend this paradigm to multimodal domains, early works primarily focused on augmenting the intermediate reasoning process with visual evidence[24, 53, 54]. While these methods successfully improve interpretability, their reasoning processes remain predominantly text-centric.

To better leverage visual information within reasoning, subsequent methods progressed by directly injecting spatial references into the reasoning chain[3, 15, 23, 45]. However, while these methods successfully make visual grounding explicit, they predominantly treat spatial references as localized, single-use annotations attached to individual reasoning steps. Consequently, entity tracking still relies on implicit natural language descriptions, leaving current models highly susceptible to cross-step reference drift. Our work addresses this limitation by introducing the reasoning anchor. By representing grounded entities as persistent, identity-bound anchors, we make intermediate references trackable across steps.

3 Method

This section presents the complete framework of RoboPIN. Section 3.1 defines the PinCoT structured reasoning representation, including the design of reasoning anchors, the identity tracking mechanism, and its natural extension to multi-view settings. Section 3.2 describes a fully automated data construction pipeline for generating high-quality PinCoT-format reasoning supervision at scale. Section 3.3 introduces the three-stage progressive training strategy and the process-supervised reward design.

3.1 Pinned Chain-of-Thought Representation

As discussed in the introduction, although existing paradigms have introduced visual coordinates into language models, these spatial references are typically isolated and lack explicit identity tracking. This often causes the reasoning trajectory to decouple from visual evidence, leading to reference drift and inconsistency across steps. To address the reasoning consistency problem in embodied reasoning, we propose Pinned Chain-of-Thought (PinCoT), whose core concept is the *reasoning anchor*.

A reasoning anchor is fundamentally different from conventional visual grounding. Traditional grounding operations such as object detection and referring expression comprehension, operate primarily at the perceptual level and are not explicitly maintained throughout multi-step reasoning. Once an object is localized, the grounding result is typically not reused as a persistent variable. In contrast, a reasoning anchor functions as a reasoning-level variable that unifies identity and grounding. Upon introduction, it binds an entity to a unique identity along with its spatial location and view

context, and persists as a variable throughout the reasoning process, forming a traceable and step-wise verifiable reasoning chain.

PinCoT encodes entities as lightweight XML tags. Two tag types are defined: `obj` for physical objects and `space` for task-relevant spatial regions. Both share a unified parameterized schema:

```
<obj name="NAME" id="ID" img_idx="VIEW" point="[x, y]">
<space name="NAME" id="ID" img_idx="VIEW" point="[x, y]">
```

Here, `name` denotes a semantic label, `id` is an explicit identity token, `img_idx` indicates the view, and `point` represents a 2D coordinate normalized to 0–1000. Formally, each reasoning anchor can be represented as a structured tuple $a = (n, i, v, p)$. This representation design tightly binds semantic meaning, identity, view provenance, and spatial localization into a unified representation.

We adopt point coordinates over bounding boxes for three reasons. First, many embodied task targets are inherently about "which location" rather than "which region" (e.g., pointing, affordance prediction, placement positions), and points naturally align with such operational targets. Second, points are more compact: they incur lower information overhead when used as anchors within a reasoning chain. Third, the combination of point and ID is better suited as a reasoning variable, analogous to a pointer in a program that refers to a precise location while carrying identity, whereas a bounding box resembles an output of a detection module. We validate this design choice through ablation experiments comparing point and bounding box representations in Section 4.4.

Building on these structured tags, PinCoT enforces an identity-aware reasoning mechanism. When an entity first appears during reasoning, it must be introduced through a complete XML tag that establishes both its spatial localization and identity. Subsequent reasoning steps reference the same entity directly through its ID, without regenerating a full tag.

This mechanism naturally extends to multi-view settings. The same entity across viewpoints shares a unified identity token, varying only in `img_idx` and `point`. With a shared identity, cross-view consistency becomes an explicit constraint within the reasoning process, thereby reducing identity drift and enabling consistent reasoning about the same entity from different observation angles.

3.2 Automated Data Construction

Training models to follow PinCoT requires aligning visual grounding and identity with the reasoning process. However, existing multimodal datasets rarely provide such unified annotations, and directly converting raw data into PinCoT format using large models tends to bypass explicit intermediate grounding steps, resulting in anchors that are not reliably grounded in visual evidence. We therefore build a fully automated three-stage pipeline organized around explicit structured intermediate representations, incorporating dedicated quality assurance throughout the process. Using this pipeline, we construct the PIN-170K reasoning dataset.

To identify relevant entities, we use Qwen3-VL-32B-Instruct [1] to analyze the current question and extract semantic descriptions of candidate entities. Since embodied task types vary significantly, a single prompting strategy cannot cover all cases. We therefore partition the data into several task categories (e.g., spatial relation reasoning, pointing, affordance prediction) and design task-specific prompts for semantic parsing. This targeted strategy produces stable semantic specifications that accurately capture target entities.

Given the parsed semantic metadata, we employ a Florence-2 [43] + SAM 2.1 [28] pipeline to obtain precise point coordinates and bounding boxes for the relevant entities. This stage retains the predicted bounding boxes, refined segmentation masks, and derived point anchors as structured grounding evidence for subsequent use. However, this pipeline is not equally reliable across all target types: for small symbolic targets such as "X" marks in images, Florence-2 + SAM 2.1 often fails to localize accurately. To address this, we introduce a reverse prediction verification step, where Florence is asked to identify what the predicted point corresponds to. Only predictions whose reverse interpretation is consistent with the original target description are retained, substantially improving the quality and reliability of the grounding annotations.

We then feed the structured grounding evidence as ground-truth information into Qwen3-VL-32B-Instruct to generate the corresponding PinCoT reasoning chains. To improve generation quality, we impose task-specific constraints for different task categories, reducing arbitrary or unsupported reasoning during generation. After obtaining PinCoT outputs, we further apply rule-based filtering to remove low-quality reasoning chains and inconsistent results, such as incorrect ID references or mismatched anchor coordinates, ensuring the overall quality of the constructed data.

Through this pipeline, multimodal datasets are progressively transformed into structured reasoning data. The entire pipeline executes fully automatically without manual annotation and scales to large volumes of data, yielding the PIN-170K reasoning dataset. Further details regarding the pipeline can be found in Appendix A.

3.3 Progressive Training Pipeline

Building on the framework and data, we start from Qwen3-VL-4B-Instruct [1] and progressively inject the capabilities through three-stage post-training: Stage 1 performs embodied domain adaptation via SFT, Stage 2 learns PinCoT structured reasoning via CoT-SFT, and Stage 3 achieves process-supervised alignment via RFT.

We train RoboPIN on a diverse corpus to improve spatial understanding, multi-view reasoning, pointing, and long-horizon embodied reasoning. Data subsets are assigned to training stages according to their supervision signals and capabilities. For Stage 2, we apply the data construction pipeline described in Section 3.2 to generate structured PinCoT samples for learning reasoning anchor introduction and the identity-aware reasoning mechanism.

Geometric and general spatial reasoning data. This subset is primarily derived from **Euclid** [16] and **Video-R1** [11]. Euclid provides structured problems involving spatial relations and geometric properties, helping the model build precise spatial reasoning capabilities. Video-R1 provides general visual understanding samples from dynamic and diverse scenes, preserving the model's broad perceptual abilities during training.

Embodied spatial reasoning and pointing data. This subset is primarily from **EmbSpatial** [10], **EO-Data** [26], **RoboVQA** [30], **EgoPlan** [4], and **Embodied-Point** [48]. EmbSpatial and EO-Data focus primarily on embodied spatial perception, driving the model to develop robust environment-level reasoning capabilities. RoboVQA

and EgoPlan emphasize task-oriented and egocentric embodied scenarios, introducing action-related reasoning and sequential decision-making processes. Furthermore, Embodied-Point covers diverse multi-task pointing scenarios, helping the model learn to generate precise point coordinates in embodied environments.

Multi-view understanding data. This subset is primarily derived from **CrossPoint** [40] and **SAT** [29], and augmented with additional multi-view data constructed from **InternData-M1** [8]. CrossPoint and SAT provide observations of the same scene under diverse camera viewpoints, enabling the model to learn consistent object representations and spatial relationships across views. InternData-M1 provides a diverse array of scenes, serving as a crucial foundation for multi-view reasoning in embodied tasks.

Detailed statistics of the training corpus, including the size of each data subset and the mixture ratios used across progressive training stages, are provided in Appendix B.

The first stage focuses on embodied domain adaptation. We finetune the model on embodied data covering object understanding, egocentric activity recognition, pointing and planning, enabling it to build foundational understanding of objects, affordances, spatial relations, and action events. A small amount of general-domain data is mixed in to mitigate catastrophic forgetting. The second stage injects PinCoT structured reasoning ability. We leverage the pipeline from Section 3.2 to generate structured, multi-step PinCoT reasoning traces for supervised learning. In this stage, the model learns two key capabilities: introducing entities through complete XML tags to establish their reasoning anchors, and maintaining consistent identity references across both multi-step reasoning chains and multiple viewpoints.

The third stage focuses on process-supervised alignment. After the first two stages, the model can generate structured reasoning chains, but may still prefer fluent yet poorly anchored outputs during inference. To counteract this tendency, we apply GRPO [32] with a composite reward function that simultaneously supervises both the reasoning process and the final answer.

For a generated response y, let $\hat{a}$ denote the parsed final answer, $\mathcal{M}(y)$ the set of entity mentions appearing in the `<think>` block, and $\mathcal{M}_{\text{valid}}(y) \subseteq \mathcal{M}(y)$ the subset whose first occurrence is introduced by a valid format and whose subsequent mentions strictly maintain cross-step identity consistency. We further denote by $\mathcal{P}_{\text{xml}}(y)$ the set of anchor points extracted from the reasoning trace. For certain specific tasks, each predicted point p_j is paired with a target grounding region $b_j^\star$.

We design a composite reward that evaluates four complementary dimensions of output quality:

$$R = \lambda_f R_{\text{format}} + \lambda_c R_{\text{consistency}} + \lambda_p R_{\text{pin}} + \lambda_a R_{\text{accuracy}}, \tag{1}$$

where $\lambda_\cdot$ are task-specific weights. When a reward term is not applicable to a particular task, its weight is set to zero.

Format reward. R_{format} evaluates whether the output structure is well-formed. The reasoning process must be enclosed within `<think>...</think>` and the final answer within the format of `<answer>...</answer>`. The `<think>` block must also contain valid reasoning anchor tags. The highest reward is assigned when both formatting and reasoning anchor constraints are satisfied, a lower reward when only the outer format is correct, and zero otherwise.

Consistency reward. $R_{\text{consistency}}$ evaluates the coherence of entity references throughout the reasoning process, measuring the proportion of entity mentions that follow the identity-aware reasoning mechanism:

$$R_{\text{consistency}} = \frac{|\mathcal{M}_{\text{valid}}(y)|}{\max(|\mathcal{M}(y)|, 1)}. \tag{2}$$

This reward encourages the model to maintain stable entity references across reasoning chains and multiple views, reducing identity drift and reference ambiguity.

Pin reward. R_{pin} evaluates the localization accuracy of reasoning anchors in the reasoning process. This term is applied only to datasets with explicit target region supervision, avoiding over-constraining the model's reasoning on other tasks:

$$R_{\text{pin}} = \frac{\left|\left\{p_j \in \mathcal{P}_{\text{xml}}(y) \mid p_j \in b_j^\star\right\}\right|}{\max(|\mathcal{P}_{\text{xml}}(y)|, 1)}. \tag{3}$$

By directly supervising the localization accuracy of intermediate anchors, the pin reward ensures that reasoning anchors during reasoning are genuinely "pinned" to the correct locations.

Accuracy reward. R_{accuracy} evaluates the correctness of the final answer. Since answer formats vary across tasks, this reward is instantiated in a task-dependent manner: for multiple-choice tasks, it is a binary indicator of exact match; for pointing tasks, it measures the proportion of predicted points falling within the target bounding boxes; for open-ended tasks, it uses normalized sentence-level BLEU [25] as a soft text-matching reward.

Overall, the reward design introduces process supervision over *how the model reasons*. R_{format} ensures well-formed outputs, $R_{\text{consistency}}$ enforces stable identity-aware reasoning mechanism, R_{pin} directly supervises reasoning anchor localization quality, and R_{accuracy} ensures final-answer correctness. Together, these four terms enforce reasoning quality from both the process and the final answer, ultimately driving the model to generate high-quality PinCoT reasoning chains. Further details regarding the training configurations can be found in Appendix C.

4 Experiments

We evaluate RoboPIN on embodied spatial reasoning (Sec. 4.1), point-level grounding (Sec. 4.2), and real-world robot execution (Sec. 4.3). We then ablate the contributions of PinCoT and the three-stage training recipe (Sec. 4.4), and analyze how process supervision improves reasoning quality and consistency (Sec. 4.5). Finally, we verify that our embodied alignment preserves general vision-language capabilities without catastrophic forgetting, with detailed results provided in Appendix D.

4.1 Embodied Cognition and Spatial Reasoning

We evaluate the embodied cognition and spatial reasoning capabilities of our model across 9 benchmarks encompassing 13 distinct capability dimensions. These include ERQA [37], CV-Bench [38], EmbSpatial [10], SAT [29], RoboSpatial [34], Robo-VQA [30], CrossPoint [40], MVRoboBench [12], and BLINK [13]. These benchmarks comprehensively cover various aspects of spatial understanding, object localization, and physical reasoning. To rigorously assess performance, we compare RoboPIN against two leading closed-source

Table 1: Performance on spatial reasoning benchmarks. Bold and underlined denote the best and second open-source results.

Model	Params	ERQA	CV-Bench	EmbSpatial	SAT	RoboSpatial	RoboVQA	CrossPoint	MV-RoboBench		BLINK			
									CrossV. Match	Traj. Sel	Multi. View	Vis. Corr.	Spat. Rel.	Rel. Depth
Closed-source models														
GPT-5	-	54.5	84.5	78.3	83.3	54.1	33.1	33.0	29.0	54.5	45.8	77.3	90.2	82.3
Gemini-2.5-Pro	-	55.7	84.6	78.7	76.7	59.9	33.9	37.1	39.5	65.5	46.6	78.5	85.3	86.3
Open-source generalist models														
Qwen3-VL	4B	41.8	85.0	77.1	70.7	59.4	41.1	29.3	24.0	31.0	51.1	88.4	84.6	85.5
InternVL3.5	8B	41.0	81.5	70.3	55.3	51.1	28.6	19.2	21.5	35.0	47.4	72.7	81.8	78.2
Open-source embodied models														
RoboBrain2.0	7B	38.5	85.8	76.3	75.3	54.2	57.5	26.0	21.0	26.0	48.1	46.5	80.4	80.7
VeBrain	7B	37.3	79.7	70.5	58.0	42.5	42.4	20.2	18.0	30.0	46.6	48.8	83.2	67.7
Mimo-Embodied	7B	**46.8**	**88.8**	76.2	54.6	61.8	**62.0**	36.8	21.1	34.0	26.3	83.2	77.6	**94.4**
Pelican-VL	7B	39.8	78.9	73.2	54.6	57.5	58.5	23.8	23.0	32.5	47.3	63.3	84.6	75.0
Embodied-R1	3B	35.2	82.7	67.4	76.3	47.4	51.8	25.2	21.5	26.5	40.6	50.0	76.9	79.8
RoboPIN	4B	44.0	85.5	**83.1**	**78.0**	**66.6**	56.1	**72.2**	**26.0**	**35.5**	**72.9**	**89.5**	**87.4**	87.1

models: GPT-5 [33] and Gemini-2.5-Pro [7], alongside seven competitive open-source vision-language models equipped with advanced spatial reasoning and grounding capabilities. These include generalist multi-modal models (Qwen3-VL [1] and InternVL3.5 [39]), as well as recent domain-specific models designed for embodied reasoning (RoboBrain2.0 [36], VeBrain [20], Mimo-Embodied [14], Pelican-VL [52], and Embodied-R1 [48]).

Results. As detailed in Table 1, RoboPIN demonstrates exceptional general spatial reasoning capabilities. It achieves the best open-source results on EmbSpatial, SAT, and RoboSpatial, demonstrating the advantage of PinCoT in tasks requiring relative object positions and spatial changes. Across the other embodied cognition benchmarks, our model consistently performs within the first tier. Furthermore, on multi-view evaluations, RoboPIN ranks first on CrossPoint, as well as the Multi-View and Visual Correspondence subsets of BLINK. This indicates that the universal identity design across views effectively facilitates stable object correspondence, enhancing the multi-view reasoning proficiency.

4.2 Embodied Pointing and Location

We next examine point-level grounding ability by evaluating on five distinct benchmarks: VABench-P [47], Where2Place [46], RefSpatial [55], RoboRefit [19], and RoboAfford [35].

Results. As shown in Table 2, RoboPIN achieves the best open-source performance on RefSpatial, RoboRefit, and RoboAfford. We note that Embodied-R1, as a model specifically designed and trained for pointing tasks, shows strong performance on VABench-P and Where2Place. Despite being a general-purpose embodied reasoning model, RoboPIN remains competitive on two benchmarks while significantly outperforming Embodied-R1 on broader spatial reasoning tasks, demonstrating a more balanced capability profile.

4.3 Real-World Robot Evaluation

We conduct zero-shot real-world evaluations using an xArm robot. The system is equipped with two RGB cameras: a static camera observing the tabletop scene and a wrist-mounted camera. Given a language instruction, the model predicts target points via visual grounding, which are then executed using the point-based motion planning pipeline from Embodied-R1 [48].

Table 2: Performance on pointing benchmarks. Bold and underlined values denote the best and second-best open-source results.

Model	Params	VABench-P	Where2Place	Ref-Spatial	Robo-Refit	Robo-Afford
Closed-source models						
GPT-5	-	31.1	39.7	21.6	44.7	30.0
Gemini-2.5-Pro	-	21.7	49.6	36.5	38.4	23.4
Open-source generalist models						
Qwen3-VL	4B	30.6	67.7	43.0	86.1	70.1
InternVL3.5	8B	23.0	34.8	16.8	30.8	31.5
Open-source embodied models						
RoboBrain2.0	7B	41.0	63.6	32.5	70.4	51.5
VeBrain	7B	1.7	12.3	0.3	32.2	2.1
Mimo-Embodied	7B	46.9	63.6	48.0	82.3	69.8
Pelican-VL	7B	14.5	57.8	37.5	74.9	63.4
Embodied-R1	3B	**66.0**	**69.5**	39.7	85.6	67.2
RoboPIN	4B	65.0	62.0	**50.5**	**89.0**	**72.8**

Table 3: Real-world robot evaluation. Each setting is evaluated over 5 independent trials.

Task	RoboBrain2.0	Mimo-Embodied	RoboPIN
Spatial	12/20	7/20	**20/20**
Multi-view	1/20	0/20	**14/20**
Long-horizon	5/20	3/20	**11/20**

Tasks. We design three types of complex manipulation tasks to evaluate spatial reasoning, multi-view understanding, and long-horizon planning:

(1) Spatial reasoning task: *"Pick up the duck in front of two toys and place it to the middle plate among plates."*

(2) Multi-view reasoning task: *"Pick up the duck behind two toys and place it to the plate in front of the two toys."*

(3) Long-horizon task: *"Pick up yellow objects located outside the plates and place into the yellow plate."*

To rigorously ensure robustness, we extensively randomize object instances, distractor placements, and overall scene layouts for each task. We conduct 20 distinct trials per setting under these varied conditions to compute the success rate.

Results. Table 3 reports the success rates of different methods, including RoboBrain2-7B and Mimo-Embodied-7B. In multi-view tasks, baseline models tend to rely solely on the static camera and often select incorrect targets, failing to utilize the wrist-view information. In contrast, our model effectively integrates multi-view

Table 4: Ablation of reasoning format and RFT. The first row indicates the Qwen3-VL-4B-Instruct baseline. All compared variants use the same training recipe and data subsets.

Reasoning Design			Performance	
Format	SFT	RFT	EmbSpatial	SAT
Baseline	×	×	77.1	70.7
Text CoT	✓	×	78.9	74.0
Text CoT	✓	✓	79.3	76.3
Coord CoT	✓	×	81.7	67.3
Coord CoT	✓	✓	83.1	74.0
Coord CoT w/ ID	✓	×	78.6	70.7
Coord CoT w/ ID	✓	✓	81.5	76.0
PinCoT	✓	×	80.3	74.0
PinCoT	✓	✓	**84.7**	**78.7**

observations and demonstrating enhanced reliability in identifying the target object compared to the baselines.

In spatial reasoning tasks, our model also achieves consistently higher accuracy, reflecting a stronger capacity for interpreting spatial relations and resolving ambiguous object configurations. For the long-horizon task, RoboPIN predicts keypoints step-by-step and outperforms other models at each step, leading to a substantially higher overall success rate. Figure 3 illustrates representative execution sequences alongside the predicted anchor points for multi-view and spatial tasks, visually confirming the model's ability to ground its reasoning in precise spatial locations.

4.4 Ablation Study

In this section, we conduct ablation studies on three key components: the reasoning format, the reward design, and the three-stage training pipeline. For the format and reward ablations, we use identical training hyperparameters on fixed training subsets to isolate the effects of reasoning format and reward design.

Performance Comparison of Reasoning Formats. We compare four spatial representations: text-only CoT (*Text CoT*), coordinate CoT (*Coord CoT*), coordinate CoT with IDs (*Coord CoT w/ ID*), and our PinCoT with point-based anchors and IDs. As shown in Table 4, explicit reasoning consistently improves over non-reasoning baselines, while the spatial representation remains crucial. PinCoT generally outperforms other formats, with larger gains after RFT. Although PinCoT underperforms *Coord CoT* under SFT on EmbSpatial, it surpasses it after RFT, suggesting that PinCoT provides a more effective reasoning interface when sufficiently optimized. Moreover, compared with *Coord CoT w/ ID*, which uses box coordinates with the same ID mechanism, PinCoT consistently performs better, confirming that point-based anchors provide a more suitable spatial representation for embodied reasoning.

Ablation of Reward Components. We further ablate the two core reward components in RFT: the Pin reward for anchor localization and the Consistency reward for entity references. As shown in Table 5, removing either component leads to a clear performance drop, confirming that both spatially accurate anchors and identity-consistent reasoning are necessary for strong embodied reasoning performance. Notably, removing the Pin reward causes a larger degradation (−2.7) than removing the Consistency reward (−2.0),

Table 5: Ablation of reward components on EmbSpatial.

Setting	EmbSpatial
RoboPIN	**84.7**
w/o Pin reward	82.0
w/o Consistency reward	82.7

Table 6: Ablation of the three-stage training recipe.

Training Stages			Avg-Performance	
SFT	CoT-SFT	RFT	Spatial	Point
✓	×	×	57.0	66.0
✓	✓	×	62.0	63.6
✓	✓	✓	**67.7**	**67.9**

Table 7: Reasoning process analysis on EmbSpatial. A-Cov: Anchor Coverage; AC: Anchor Correctness; CS-IDCov: Cross-Step identity Coverage.

Model	A-Cov (%)	AC (%)	CS-IDCov (%)
RoboPIN	**100.0**	**89.1**	**99.9**
w/o RFT	96.3	87.0	18.8

suggesting that anchor localization quality is the more fundamental bottleneck for embodied reasoning.

Performance Comparison of the Three-Stage Training. We further analyze how model performance progresses across different training stages based on average scores over embodied reasoning and pointing tasks. As shown in Table 6, embodied reasoning improves progressively as additional stages are introduced: CoT-SFT yields a +5.0 gain over SFT alone, and RFT provides a further +5.7 improvement. This suggests that our three-stage recipe primarily strengthens high-level embodied spatial reasoning, while effectively preserving the fundamental point-level localization capabilities.

4.5 Reasoning Process Analysis

To understand why RoboPIN achieves superior performance, we analyze the intermediate reasoning process on EmbSpatial (which provides ground-truth bounding boxes) and CrossPoint (for cross-view evaluation).

Anchor Quality and Identity Consistency. As shown in Table 7, RoboPIN achieves full anchor coverage (100.0%) and high anchor correctness (89.1%), both degrading without RFT. The key difference lies in identity tracking: without RFT, the model rarely maintains ID-based reasoning (18.8%) despite accurate anchors, while RFT achieves 99.9% cross-step coverage. This shows that process supervision enforces stable identity-aware reasoning beyond improving anchor quality. Figure 4 further shows that identity-aware tracking removes ambiguous references and improves cross-step and cross-view consistency.

Error Propagation from Reasoning to Answer. As shown in Table 8, samples with faulty anchors exhibit roughly 2.2× higher

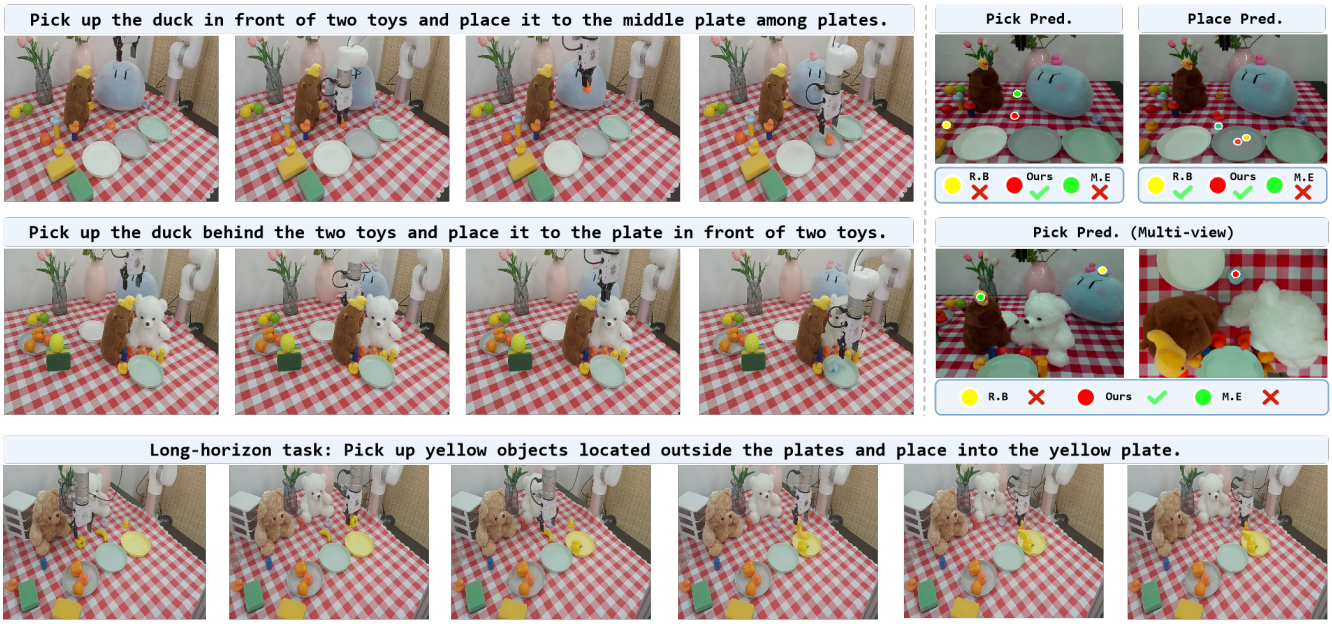


Figure 3: Real-world robot visualization. Left: robot execution. Right: predicted anchor points during manipulation. R.B and M.E denote Robobrain2 and Mimo-Embodied.

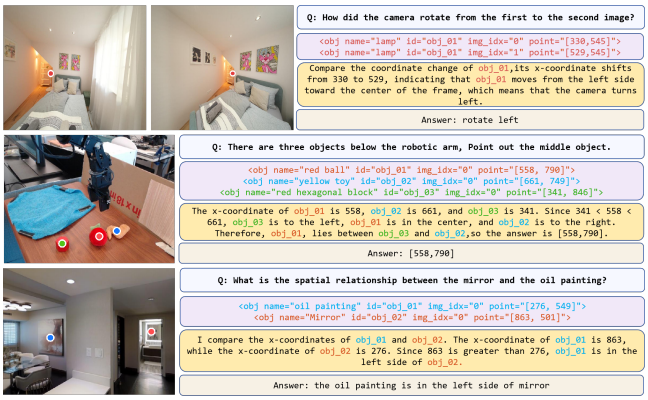


Figure 4: Qualitative analysis of grounded reasoning.

Table 8: Error propagation on EmbSpatial.

Model	Answer Error Rate (%)		Risk Ratio
	w/ Faulty Anchor	w/ Correct Anchor	
RoboPIN	29.2	13.2	2.2

answer-error rates than those with correct anchors, showing that reliable spatial grounding is critical for robust reasoning.

Cross-view Consistency. We evaluate the Pointing and Judgement subsets of CrossPoint, measuring overall accuracy (*Acc*), shared-ID transfer rate across views (*ID-Trans*), and conditional accuracy given successful transfer (*Trans-Acc*). As shown in Table 9, RoboPIN consistently maintains shared-identity transfer across views (86.3%),

Table 9: Cross-view reasoning consistency on CrossPoint. Acc: Overall Accuracy; ID-Trans: Shared-ID Transfer Rate; Trans-Acc: Accuracy given successful ID transfer.

Model	Acc (%)	ID-Trans (%)	Trans-Acc (%)
RoboPIN	**71.1**	**86.3**	**72.1**
w/o RFT	62.5	60.6	61.1

and when this transfer succeeds, predictions become highly reliable (72.1%). Removing RFT substantially weakens this behavior (ID-Trans drops to 60.6%), accompanied by a clear performance decline. These findings confirm that process supervision is the critical driver enforcing explicit cross-view correspondence.

5 Conclusion

We present PinCoT, a structured reasoning paradigm that anchors each reasoning step to visual evidence by unifying semantic identity and spatial localization into structured reasoning variables, enabling persistent entity tracking across steps and viewpoints. Based on an automated data pipeline and three-stage training, we develop RoboPIN, where RFT further improves reasoning-process and answer consistency. Across 14 embodied VLM benchmarks, 4B-parameter RoboPIN achieves a 12% average improvement over the strongest 7B baseline Mimo-Embodied. Our results demonstrate that explicitly grounding reasoning in visual evidence is an effective approach for improving embodied VLM spatial reasoning. Future work will explore larger-scale models and data, as well as extending PinCoT to 3D localization and long-horizon planning.

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